come together in the fertilized cell is extremely small. No radiation induced hereditary effects have been observed in mammals. Irradiation of the sexual organs of 2000 carefully selected mice (~2 Sv each) for 19 generations (in total 38 000 mice, all inbred) showed no genetic changes (Lüning 1970). A thorough investigation of the 75 000 children born to parents who were exposed to the A-bomb irradiation in 1945 have not shown any increase in the frequency of hereditary diseases (or cancer). The 35 000 children born to parents of which at least one had been exposed to $\leq$3 Sv (average exposure 0.2 Sv) showed no genetic differences (analyzing still births, birth weight, congenital abnormalities, infant mortality, childhood mortality, leukemia or sex ratio) from children born to non-irradiated parents within statistical uncertainty.

Large doses to plants have caused mutations which have either improved the quality of the species or produced effects which are disadvantageous to the species, but desirable to society. Although irradiation of plant seeds results in a ratio of about 1000 to 1 of the harmful to the advantageous mutations, by cultivating those few plants showing improvement in properties, new plant variations have been obtained. This has resulted in species of grains and legumes which have stronger stocks, higher yields, and improved resistance to mold and to fungi. In northern countries such as Sweden, most of the grain that is grown today is radiation-produced species possessing much greater cold resistance.

As a conclusion one might say that genetic effects in higher mammals from radiation has not been observed. After the Chernobyl accident thorough studies of fast reproducing mammals such as rats have shown no genetic effects.

15.9. RADIOMIMETIC SUBSTANCES

Many chemical substances when administrated to the body show the same effects as irradiation. Because cancer caused by radiation has been investigated more than cancer induced by chemicals, such substances are called *radiomimetic* (mimetic = imitative). In order to qualify as a radiomimetic agent, the substance must do the following: stop cell division, stop tumor formation, produce chromosome aberrations, cause mutations, kill lymphocytes, and be carcinogenic. Chemical substances which meet a few, but not all of these requirements are not considered radiomimetic. The effects depend on the concentration of the substance; e.g. cell division is interrupted by many radiomimetic substances at concentrations $\leq 10^{-5}$ mole l^{-1}.

Typical radiomimetic substances are organic peroxides (e.g. ethylene oxide), ethylene di-imine, mustard gas and derivatives, aliphatic dichloro-amines, etc. These compounds or chemical groups occur in many familiar materials such as tobacco smoke. The effect of a certain amount of a radiomimetic substance can be calculated to correspond to a radiation dose. Thus, smoking a pipe of tobacco corresponds to an average radiation dose of about 0.04 mSv.

15.10. RADIATION BACKGROUND

Through life, everyone is constantly exposed to ionizing radiation from a variety of sources. Table 15.8 gives radiation doses which are representative for a large number of countries with the exception of extreme conditions like cosmic radiation at high altitude (e.g La Paz at altitude 3900 m) and locations on monazite sand (e.g. Ramsar, Iran, Kerala, India, and Esperito Santo, Brazil). Table 15.9 gives the relative contribution of sources of radiation to the UK population. The values are representative for most countries in the northern hemisphere. Within the same country, the total background dose commonly varies within a factor of 3–5 between lowest and highest areas. Brick and concrete houses contain varying amounts of uranium and/or radium, depending on the source of the building material. Because of radioactive soil, all foods contain some natural radioactivity. An approximate average value in northern Europe is <0.3 Bq/kg from U-decay products in food or beverage, while vegetables in the monazite areas in India (e.g. roots) have been found to have >10 Bq/kg.

Presently great concern is directed towards people living in dwellings with high radon concentration ($\geq$400 Bq/m^3) because the radon and daughters are assumed to

Table 15.8 Effective dose (normal → elevated value) in mSv/y (UNSCEAR 1993) or average annual dose values (Nias 1990)

Source	Value
Cosmic rays	0.3 → 2.0 (LaPaz) mSv/y
Terrestrial, external	0.3 → 4.3 (Kerala) mSv/y
Intake	0.052 (ingestion), 0.01 (inhalation) mSv/y
Radionuclides in body[†]	0.23 → 0.6 mSv/y
Rn in water supplies and air:	see §13.6
Rn in body	1.3 → 10 mSv/y
Buildings[‡]: wood/gypsum	0.18 mSv/y (K 150, Ra 20, Th 5 Bq/kg)
" : typical masonry	0.7 " (K 500, Ra 50, Th 50 Bq/kg)
" : alum shale concrete	5.9 " (K 770, Ra 1300, Th 67 Bq/kg)
Nuclear power plant: to critical group (Sweden)	<0.1 mSv/y
X-ray investigation:	5–8 mGy
barium meal (intestine)	
intravenous urography	4–5 "
abdomen and lung	~1 "
mean gonadal dose (abdomen),	<0.2 "
genetically significant dose	0.1–0.5 "
dental	0.02 "
Total average	2–4 mSv/y

[†]Except Rn.
[‡]Staying indoors at 8760 h/y; masonry building in temperate climate.